

CHAPTER 7

AGE-RELATED DIFFERENCES AND THEIR IMPLICATIONS FOR RESISTANCE TRAINING

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After completing this chapter, you will be able to

- understand the merits of adopting a long-term approach to the development of athleticism for child and adolescent populations;
- evaluate the evidence regarding the safety, effectiveness, and importance of resistance exercise for youth populations;
- describe the effects of aging on musculoskeletal health and performance;
- explore the trainability of older adults and the benefits of exercise, including resistance training in aging populations; and
- explain why adaptations to resistance exercise vary greatly among these distinct populations.

Resistance exercise has been proven to be a safe and effective method of conditioning for individuals with various needs, goals, and abilities. Resistance training is also recognized as a fundamental

component of long-term athletic development that helps to support lifelong participation in sport and recreational physical activity from childhood to the more senior years of life (137). In light of its many benefits for all individuals, resistance exercise for youth (137) and older people (76) has received increasing public and medical attention. However, when designing and evaluating resistance training programs, strength and conditioning professionals need to understand the implications of age-related differences in body composition, muscular performance, trainability, and injury risk to facilitate individualized exercise prescription.

For the purposes of this chapter, **resistance exercise** is defined as a specialized method of conditioning whereby an individual is working against a wide range of resistive loads to enhance health, fitness, and performance. This term should be distinguished from the sport of weightlifting, in which individuals attempt to lift maximal amounts of weight in competition, specifically in the clean and jerk and snatch exercises. The term **childhood** refers to a period of life before the development of secondary sex characteristics (e.g., pubic hair and reproductive organs), and the term **adolescence** refers to the period between childhood and **adulthood**. For ease of discussion, the term **youth** or **young athlete** refers to both children and adolescents. The terms **older** and **senior** include men and women over 65 years of age. **Athleticism** reflects the ability to repeatedly execute a variety of movements with precision and confidence in a range of environments, which require appropriate levels of motor skill competency, balance, coordination, strength, power, speed, agility, and endurance (137). The development of athleticism is important for everyone and should not be reserved as an outcome goal only for those involved in organized sport.

Youth Populations

The National Strength and Conditioning Association considers **long-term athletic development** as the development of athleticism over time, with an aim to enhance physical fitness, reduce injury risk, and improve mental and physical health and well-being (137). The concept of regularly participating in activities that enhance athleticism is relevant for people of all ages. However, over the past two decades, there has been a growing focus on long-term athletic development programs, particularly for youth populations (137). Athletic development models that encompass multiple modes of training have been proposed previously in the literature. For example, Lloyd and Oliver (141) proposed the youth physical development (YPD) model, which provides training guidelines for males and females ([figure 7.1a](#) and 7.1b). The YPD model is based on the notion that all components of fitness are trainable at all stages of development, and therefore training programs should be multifaceted in nature to develop multiple dimensions of athleticism. Crucially, the model promotes a consistent focus on the simultaneous development of muscle strength and motor skill competency, which provide the foundations for overall athleticism (137). The model also supports children engaging in some form of physical activity and preparatory conditioning early in life to take advantage of the heightened neuroplasticity associated with early childhood development (88). Similarly, training prescription should transition from low structure and movement exploration early in a child's development to a more structured and targeted approach as individuals accumulate training experience and move toward later stages of development.

However, several authors have also established more detailed development models for individual components of fitness, such as agility (144), speed (176), power (155), and endurance (104), and specific modes of training, such as motor skills (194), resistance training (91), plyometrics (140), and weightlifting (168). While each

model focuses on bespoke elements of athletic development, several notable commonalities exist between models: (a) Children should be exposed to a wide range of training activities that target motor skill development and muscular strength, (b) training should begin simple and become more advanced as children progress through the various stages of development, and (c) training should be generic to start with and then become more specialized over time as workloads and training exposure increase.

Collectively, strength and conditioning programs effectively promote the development of athleticism in both children and adolescents (141). In particular, leading international consensus (17, 137, 139, 215) and seminal meta-analytical data (132, 134) have indicated the effectiveness of resistance training in enhancing multiple components of fitness in youth populations. While the effectiveness of resistance training in youth is unequivocal, it is important for strength and conditioning professionals to understand how the fundamental principles of growth, maturation, and development can affect training adaptations, confound interpretation of testing or monitoring data, and can influence the development and evaluation of safe and effective resistance training programs. Because youth sports training has become more intense and complex (21), anatomical, physiological, and psychosocial factors that may be associated with acute and chronic injury also need to be considered.

The Growing Child

In this section, the terms **growth**, **development**, and **maturation** are used to describe changes that occur in the body throughout life. The term *growth* refers to an increase in body size or a particular body part; *development* describes the natural progression from prenatal life to adulthood; and *maturation* is defined as progress toward a fully mature state, which varies in timing, tempo, and magnitude between different bodily systems (142). **Puberty** refers to a period of time in which secondary sex characteristics develop

and a child transitions to young adulthood. During puberty, changes also occur in body composition and the performance of physical skills, with such changes varying markedly between individuals.

Chronological age (years)	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21+
Age periods	Early childhood			Middle childhood							Adolescence							Adulthood		
Growth rate	Rapid growth			↔ Steady growth			↔ Adolescence spurt			↔ Decline in growth rate										
Maturational status	Years pre-PHV										← PHV		→ Years post-PHV							
Training adaptation	Predominantly neural (age-related)										↔ Combination of neural and hormonal (maturity-related)									
Physical qualities	FMS		FMS			FMS			FMS											
	SSS		SSS			SSS			SSS											
	Mobility		Mobility							Mobility										
	Agility		Agility							Agility			Agility							
	Speed		Speed							Speed			Speed							
	Power		Power							Power			Power							
	Strength		Strength							Strength			Strength							
	Hypertrophy										Hypertrophy		Hypertrophy						Hypertrophy	
	Endurance and MC		Endurance and MC							Endurance and MC			Endurance and MC							
Training structure	Unstructured			Low structure					Moderate structure			High structure			Very high structure					

a

Chronological age (years)	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21+		
Age periods	Early childhood			Middle childhood							Adolescence							Adulthood				
Growth rate	Rapid growth			↔		Steady growth			↔		Adolescence spurt				↔		Decline in growth rate					
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Training adaptation	Predominantly neural (age-related)										↔		Combination of neural and hormonal (maturity-related)									
Physical qualities	FMS		FMS				FMS		FMS													
	SSS		SSS				SSS		SSS													
	Mobility		Mobility							Mobility												
	Agility		Agility							Agility				Agility								
	Speed		Speed							Speed				Speed								
	Power		Power							Power				Power								
	Strength		Strength							Strength				Strength								
	Hypertrophy										Hypertrophy		Hypertrophy						Hypertrophy			
	Endurance and MC			Endurance and MC							Endurance and MC				Endurance and MC							
Training structure	Unstructured			Low structure							Moderate structure			High structure			Very high structure					

b

FIGURE 7.1 The youth physical development model for (a) males and (b) females. Font size refers to importance; light blue and light red boxes refer to preadolescent periods of adaptation; dark blue and dark red boxes refer to adolescent periods of adaptation. FMS: fundamental movement skills; MC: metabolic conditioning; PHV: peak height velocity; SSS: sport-specific skills; YPD: youth physical development.

Adapted by permission from R.S. Lloyd and J.L. Oliver, "The Youth Physical Development Model: A New Approach to Long-Term Athletic Development," *Strength and Conditioning Journal* 34, no. 3 (2012): 61-72.

Chronological Age Versus Biological Maturation

Because of considerable variation in the rates of growth and development, it is not particularly accurate to define a stage of maturation or development by age in months or years, which is known as **chronological age**. Children do not grow at a constant rate, and there are substantial interindividual differences in physical development at any given chronological age. For example, a group of 14-year-old children can have a height difference as great as 9 inches (23 cm) and a weight difference up to 40 pounds (18 kg), while an 11-year-old girl may be taller and more physically skilled than an 11-year-old boy. These differences correspond to variations in the timing, tempo, and magnitude of maturation (142). The onset of puberty can vary from 8 to 13 years in girls and from 9 to 15 years in boys, with girls beginning puberty approximately two years before boys.

Given the limitations associated with chronological age, development can be better assessed by **biological maturation**, which can be measured in terms of skeletal age, sexual maturation, or somatic (physique) maturation. The assessment of maturation in children is important for several reasons. Maturity assessment can be used to evaluate growth and development patterns in children. In addition, since the degree of maturation is related to measures of fitness, including muscular strength and motor skill performance (120), techniques used to assess maturation can help ensure that children are more fairly matched for fitness testing and athletic competition, as opposed to being grouped according to chronological age. In adequately nourished children there is no scientific evidence

that physical training delays or accelerates growth or maturation in boys and girls (70). Further, the osteogenic benefits of physical activity, specifically weight-bearing activities that generate compressive forces, are essential for skeletal remodeling and growth (228).

The gold standard for determining biological maturation is skeletal age assessment. This technique involves trained radiographers comparing X-rays or radiographs of a child against standard reference radiographs to determine the extent of ossification of the bones of the left wrist (93, 201, 217-219). *Ossification* refers to the process of laying down new bone material by cells called *osteoblasts*. While skeletal age provides the most accurate and reliable form of assessing maturity, concerns with cost, specialist equipment, time constraints, and the need for specific radiographic expertise make the method unrealistic for most practitioners working with youth. Furthermore, strength and conditioning professionals are not qualified to determine sexual age as an indicator of maturation, given the invasive nature of this assessment.

For most practitioners, the most realistic and feasible means of estimating biological age is from somatic assessments (142). Somatic age reflects the degree of growth in overall stature or smaller subdimensions of the body (e.g., limb length). Measures of growth (e.g., standing height and seated height) are relatively easy to collect, are noninvasive, and require minimal equipment. Techniques available to the practitioner include longitudinal growth curve analysis, percentages and predictions of final adult height (125), and the prediction of age from **peak height velocity (PHV)** (158), which is defined as the age at maximum rate of growth during the pubertal growth spurt (59) with the mean age, on average, being around 13.8 years in boys and 11.9 year in girls (10). While the percentage of predicted adult height (125) and the maturity offset (158) have both been used to assign youth into maturity categories, longitudinal data indicate that the percentage of predicted adult height might be more accurate at predicting

observed age at PHV (184); this appears to be a valid alternative for classifying maturity status when compared to longitudinal growth curve analysis (166). Using age, height, weight, and mid-parental height, a child's eventual adult height can be estimated, and their current height can then be determined as a percentage of predicted adult height (125). For example, a 13-year-old boy who is 165 centimeters (5 ft 5 in) tall, 60 kilograms (132 lb) in weight and whose mother and father are 160 centimeters (5 ft 3 in) and 185 centimeters (6 ft 1 in), respectively, would be estimated at approximately 90% of their predicted adult height. In some cases, it may be appropriate to measure somatic growth every three months (142). More frequent measures could be taken to create a rolling average and a smoother growth curve; however, this is not always feasible. For most settings, taking measurements every three months is likely more practical. This interval should be long enough to prevent practitioners from misinterpreting small growth changes caused by measurement inconsistencies. However, as a child nears their peak height velocity (PHV), it may be wise for practitioners to measure more frequently, such as monthly. This increased frequency is due to the rapid growth rate associated with this developmental stage.

The adolescent growth spurt ensues approximately 88% to 95% of predicted adult height, and longitudinal data indicate that PHV occurs at approximately 91% to 92% of predicted adult height (216). The risk of injury can be elevated in individuals who are around the time of PHV, and especially those who are also experiencing high rates of growth during this period (116, 165). Therefore, in the earlier example of the 13-year-old, his or her training program might need to be modified (and potentially more frequent measures of height taken) as the child approaches PHV in the event of disrupted coordination and control and heightened injury risk. Alterations in center of mass, muscle imbalances, and the relative tightening of the muscle-tendon units spanning rapidly growing bones are potential risk factors for overuse injuries during

the pubertal growth spurt (170, 225). Strength and conditioning professionals may need to adjust training programs (e.g., reinforce high-quality movement patterns, target flexibility restrictions, correct muscle imbalances, or decrease the volume or intensity [or both] of training) during these periods of rapid growth. If a young athlete complains of pain or discomfort during a growth spurt, the strength and conditioning professional should also consider the possibility of an overuse injury rather than labeling these complaints “growing pains” and, in consultation with the child’s parents or guardians, should refer the child to a medical practitioner.

Notwithstanding the influence of growth and maturation, training programs should be tailored to accommodate individual differences in ability and previous training experience, particularly when prescribing exercise for children in the weight room. It might be suitable for an early-maturing, experienced, and technically competent 13-year-old girl to be exposed to more advanced forms of training, whereas a late-maturing 16-year-old boy with no training history and low levels of technical competency may not be ready for the demands of heavy resistance exercise and require more introductory training approaches. Therefore, while a given training program will have an intended adaptation (e.g., strength and power development), a child’s **technical competency** (i.e., how well they move) in addition to their **training age** (i.e., the length of time the child has consistently followed a formalized and supervised resistance training program) and their resistance training skill literacy (i.e., how children think, connect, and feel about resistance training) should be key considerations in order to develop appropriate training programs (61, 137, 142). Strength and conditioning professionals must recognize all these variables and should individualize training program design based on each child’s technical competency, training age, and maturity level. When creating any youth resistance training program, strength and conditioning professionals should also take into consideration the unique psychosocial needs of each individual child and then design, implement, and revise programs according to

these needs. For example, a strength and conditioning professional will typically need to use different interpersonal skills to coach an inexperienced child with low technical competency and low levels of self-confidence, versus an experienced and highly competent adolescent who simply lacks motivation.

Two children of the same chronological age can differ biologically by up to five years, which underlines the importance of accounting for maturation in the design of individualized training programs.

Muscle and Bone Growth

As children grow, muscle mass steadily increases. At birth, approximately 25% of a child's body weight is muscle mass, and by adulthood this percentage increases to about 40% (146). During puberty, marked increases in hormonal concentrations (e.g., testosterone, growth hormone, and insulin-like growth factor) in boys result in a discernible increase in muscle mass and widening of the shoulders, whereas in girls an increase in estrogen production causes increased body fat deposition, breast development, and widening of the hips (59). Although muscle mass in girls continues to increase during adolescence, the increase occurs at a lower rate than in boys due to differences in their hormonal profiles (146). Throughout this time, the increase in muscle mass in both sexes is due to the hypertrophy of individual muscle fibers rather than hyperplasia (146). Peak muscle mass occurs between the ages of 16 and 20 years in females and between 18 and 25 years in males unless affected by exercise, diet, or both (146).

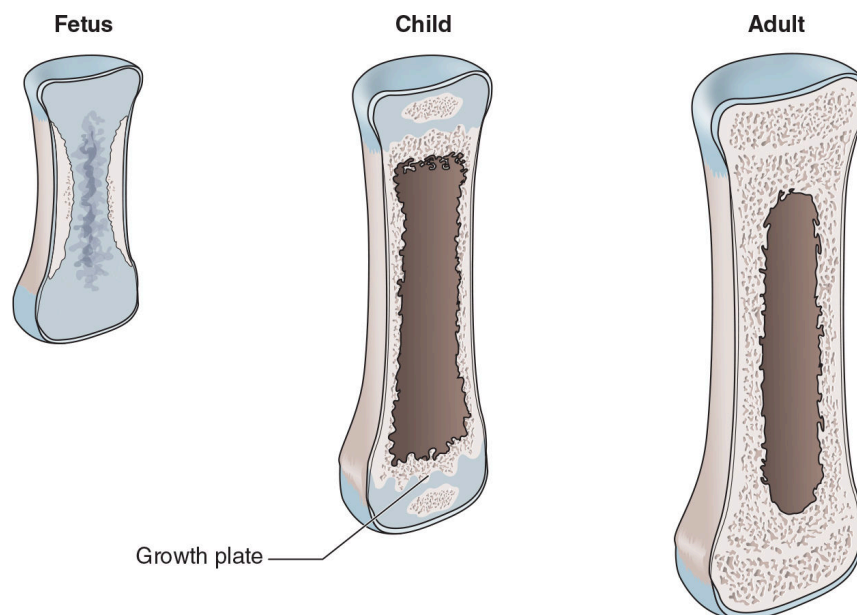


FIGURE 7.2 Bone formation, which takes place as a result of growth and development.

Most bone formation occurs in the **diaphysis** (primary ossification center), which is the central shaft of a long bone, and in the **growth cartilage** (secondary ossification center), which is located at three sites in the child: the epiphyseal (growth) plate, the joint surface, and the **apophyseal** insertions of muscle–tendon units. When the epiphyseal plate becomes completely ossified, the long bones stop growing ([figure 7.2](#)). Although bones typically begin to fuse during early adolescence, girls generally achieve full bone maturity about two to three years before boys. The actual age varies considerably, but most bones are fused by the early 20s.

A particular concern in children is the vulnerability of the growth cartilage to trauma and overuse (106). Injuries to the growth cartilage may disrupt the bone’s blood and nutrient supply and result in permanent growth disturbances (e.g., skeletal undergrowth, skeletal overgrowth, or malalignment of bone). Trauma from falls or excessive repetitive stress that may result in a ligament tear in an adult may produce an **epiphyseal plate** fracture in a child. Because the peak incidence of epiphyseal plate fractures in children occurs at around the time of PHV, it seems that a preadolescent child may be

at less risk for an epiphyseal plate fracture than an adolescent child experiencing the growth spurt (156). It has been suggested that the epiphyseal plates of younger children may be stronger and more resistant to shearing-type forces, which may be the cause of injuries to the growth cartilage (156).

Growth cartilage in children is located at the epiphyseal plate, the joint surface, and the apophyseal insertions. Damage to the growth cartilage may impair the growth and development of the affected bone. However, the risk of such damage can be reduced with appropriate exercise technique, sensible progression of training loads, and suitable instruction by qualified strength and conditioning professionals.

Developmental Changes in Muscular Strength

As muscle mass increases throughout preadolescence and adolescence, so does muscular strength. In fact, the growth curves for strength are similar to those for body mass (59). In boys, peak gains in strength typically occur about 1.2 years after PHV and 0.8 years after peak weight velocity, with body weight being the clearer indicator (146). This reflects the adage that children “stretch out before they fill out”; that is, they experience growth in height before gaining muscle mass. This pattern also suggests that during periods of rapid growth, muscle increases first in mass and later in its ability to express and attenuate high levels of force. This is reflected by meta-analytical data that show adolescents were able to attain increases in absolute muscular strength that were nearly 50% greater than children (15). In girls, peak gains in strength also typically occur after PHV, although there is more individual variation in the relationship of strength to height and body weight for girls than for boys (146). Although the strength of boys and girls is essentially equal during preadolescence, hormonal differences during puberty are responsible for acceleration in the strength gains of boys

and a general plateau in the natural strength development of girls during adolescence (139, 146). On average, peak strength is usually attained by age 20 in untrained women and between the ages of 20 and 30 in untrained men (146).

An important factor related to the expression of muscular strength in children is the development of the nervous system. If myelination of nerve fibers (motor neurons) is absent or incomplete, fast reactions and skilled movements cannot be successfully performed, and high levels of strength and power are impossible. As the nervous system continues to develop with age, children improve their performance in skills that require coordination, balance, agility, speed, strength, and power. Since the myelination of many motor neurons is incomplete until sexual maturation, children should not be expected to respond to training in the same way or to reach the same skill level as adults until they reach full neural maturity (127). However, children are well placed for skill development given the heightened neuroplasticity associated with childhood. This stage of development involves the process of pruning and strengthening of the synaptic pathways (88), which, with the appropriate environment and activities, enable children to refine motor control strategies and enhance skill acquisition. International consensus proposes that all youth should be exposed to training programs that provide an opportunity to develop a broad range of fitness components to build well-rounded youth (17, 137). However, given that childhood is a time when the corticospinal tissue is plastic, and that neuromuscular coordination and force production are governed by neural activation and control, children should primarily focus on the development of motor skill competency and muscular strength because these qualities will serve as foundations of athleticism and a lifetime of moderate to vigorous physical activity (137). Crucially, coordination and muscular strength should not be viewed as distinct qualities but as synergistic components of motor skill performance (30). Therefore, practitioners should seek to train these qualities

concomitantly instead of developing them in a hierarchical, sequential manner.

Because physiological functions are more closely related to biological age than to chronological age, at any given time an early-maturing child will likely have an advantage in measures of athleticism (e.g., absolute strength) if compared against a later-maturing child of the same sex with less muscle mass. Clearly, physical differences in body proportions can affect the execution of resistance exercises. For example, short arms and a large chest cavity offer a biomechanical advantage in upper body pressing exercises, whereas conversely, long legs and a long torso are a disadvantage in squatting movements. These anthropometric factors have implications for strength and conditioning professionals who are attempting to standardize fitness tests or develop a resistance training program for a group of boys and girls who vary greatly in physical size. It should be noted that whether in a testing or training environment, the practitioner should use child-size resistance machines or exercises using body weight, medicine balls, elastic bands, dumbbells, or barbells. In situations where practitioners are coaching groups of young athletes, they typically will need to account for variation in technical competency, maturity status, and training experience to provide a training session that is individualized wherever possible. When coaching large groups of children with low coach-to-athlete ratios, individualizing one or two additional exercises per session (based on results from simple field-based testing) might appropriately account for individualization. Similarly, training sessions that pursue a common goal (i.e., movement skill development and muscular strength) but adopt differentiating levels (e.g., technical difficulty, intensity, range of motion, movement speed) can also enable practitioners to individualize training based on the athletes' ability.

The reasons for individualized training programs should be explained to all participants, and special encouragement should be offered to late maturers who may be smaller and weaker than

chronological-age peers with more advanced biological maturity. Although late maturers tend to eventually catch up to the early maturers as they proceed through adolescence, young athletes should realize that many factors, including motivation, coaching, and innate ability, contribute to success in sport. Importantly, when coaching youth populations, especially young children (e.g., 5-7 years of age), training environments should attempt to foster task-oriented views of success (e.g., application of effort, learning something new) rather than an ego-oriented view (e.g., simply winning or beating others). While not an exhaustive list, previous literature has noted that athletic development programs should seek to cultivate a growth mindset, self-determined motivation, perceived competence, confidence, and resilience as key psychological attributes in young athletes (175).

Youth Resistance Training

Clinicians, coaches, and exercise scientists now agree that resistance exercise is a safe and effective method of conditioning for children (13, 56, 65, 71, 127, 139, 215). An increasing number of boys and girls are participating in resistance training activities, and major sports medicine organizations support children's participation in a range of resistance exercise modes, provided that the programs are appropriately designed and supervised by qualified professionals (21, 56, 139, 215). National standards and grade-level outcomes for physical education include guidelines and recommendations that recognize the importance of fitness activities that enhance muscle and bone strength (52). Perhaps most notably, the World Health Organization stipulates that children should engage in muscle- and bone-strengthening activities three times per week as part of their physical activity recommendations (235).

It is imperative that strength and conditioning professionals remember that children are not miniature adults. No matter how big or strong a child is, children are physically, psychologically, and socially less mature and are often experiencing training activities for

the very first time. Children should begin resistance training at a level that is commensurate with their physical and technical abilities, maturity level, and individual goals (141). Adult programs and training philosophies should not be superimposed on younger populations. In such instances, the intensity and volume of training are often too severe, and the recovery between training sessions is inadequate to enable training adaptation to take place. When introducing children to resistance training activities, it is always better to underestimate their physical abilities and gradually increase the volume and intensity of training than to exceed their abilities and risk injury or long-term negative health outcomes.

Responsiveness to Resistance Training in Children

A plethora of research now shows both children and adolescents can make meaningful improvements in a wide range of athletic qualities in response to different modes of resistance training (59, 141). Much of the early controversy surrounding youth resistance training stemmed from the issue of children's trainability (i.e., children's responsiveness to the stimulus of resistance exercise) due to early studies failing to demonstrate a strength increase in preadolescents who participated in a resistance training program (49). Although the lack of significant findings in these studies could be explained by methodological shortcomings, such as short study duration or inadequate training volume or intensity, the results from these reports are sometimes cited as proof that resistance training is ineffective for children. As previously discussed, muscular strength normally increases from childhood and throughout the teenage years; thus, a more appropriate conclusion from these reports may be that training-induced gains from a short-duration, low-volume, and low-intensity training program are not distinguishable from gains attributable to normal growth and maturation.

Other investigations have clearly demonstrated that boys and girls can increase muscular strength above and beyond that accompanying growth and maturation alone, provided that the

intensity and volume of training are adequate (48, 63, 135, 138, 197). Children as young as age five have benefited from developmentally appropriate resistance training (3, 121), and a variety of training modalities have proven to be effective (139). While gains in a maximum strength range of approximately 10% to 90% have been reported in the literature (15), strength gains of roughly 30% to 40% are typically observed in untrained preadolescent children following short-term (8-20 weeks) resistance training programs (56, 139). However, following the initial adaptation period, the rate of change in strength gains will be attenuated as youth adapt to the training program, thus highlighting the need for continued, progressive training. Meta-analytical data have shown that resistance training interventions realize larger gains in muscle strength and power when programs exceed 23 weeks in duration (134). Longer-term training programs afford more opportunities for children to learn and develop skills but also enable children to be exposed to higher intensities of training in a safe and progressive manner. Lesinski and colleagues (134) also indicated that strength gains were greater when young athletes were exposed to higher relative intensities ($>80\%$ 1RM). Similar notions apply to other fitness components, with data showing larger gains in jump height in response to moderate- to high-intensity resistance training (212) and greater gains in peak oxygen uptake ($\dot{V}O_{2peak}$) when endurance training is prescribed at $\geq 85\%$ HRmax (5). Clearly, training intensity is a critical prescription variable for strength and conditioning coaches to manipulate; however, in accordance with the principle of progressive overload, any increases in intensity should be applied in a progressive and gradual manner, with periodized fluctuations in volumes and intensities over time (59).

It is crucial that practitioners ensure that training intensity is never increased at the expense of technical competency or a child's psychosocial maturity. Programs should always remain designed in accordance with the

specific needs, abilities, and aspirations of the individual.

Children who participate in resistance training programs are likely to undergo periods of reduced training or inactivity due to program design factors, extended travel plans, busy schedules, injury, involvement in multiple sports, or decreased motivation. This temporary reduction or withdrawal of the training stimulus is called *detraining*. In children, unlike adults, the evaluation of strength changes during the detraining phase is complicated by the growth-related strength increases during the same period. Nevertheless, data indicate that training-induced strength gains in children are impermanent and tend to return to untrained control group values during the detraining period (55, 69). In one report, participation in physical education classes and organized sports throughout an eight-week detraining period did not maintain the preadolescents' training-induced strength gains (69). In another study comparing the effects on children of one and two days per week of resistance training, participants who resistance trained only once per week averaged 67% of the strength gains of participants who resistance trained twice per week (63). While increased training frequency is related to a greater strength training effect in youth (15), this finding must be considered in light of lifestyle factors (e.g., nutrition and sleep) and other commitments that youth or young athletes may have (e.g., competitive playing schedule, schoolwork, time to interact with peers). Though the precise mechanisms responsible for the detraining response remain unclear, it seems likely that changes in neuromuscular functioning are at least partly responsible. Further, the detraining effect may not be homogenous, with different neuromuscular qualities in young children showing different responses following the cessation of a training program (55, 172).

The potential trainability of children and adolescents is evident from research studies that have adopted long-term training intervention periods. For example, a series of studies examined the

effects of a two-year strength training program on strength (123), speed (208), and change-of-direction speed (124) in youth soccer players in Germany. Using a linear periodized training approach (i.e., four weeks of technique block, eight weeks of hypertrophy emphasis block, and eight weeks of strength emphasis, all of which repeated twice), young athletes ranging from 11 to 19 years of age were able to make substantial increases in squat strength (123), which also translated to improvements in speed (208) and change-of-direction-speed (124) performance.

Notwithstanding the clear benefits of resistance training, combined resistance training (i.e., strength training and plyometrics) has been shown to be more effective than strength training in isolation (193). Previous research has investigated the effects of 28 weeks of combined resistance training versus combined resistance training with weightlifting on physical fitness (187) and established lower limb injury risk factors (188) in adolescent boys. Notably, both training groups were able to realize improvements in physical fitness (i.e., strength, power, speed), movement competency (i.e., resistance training skills battery quotient), and lower limb jump-landing mechanics (i.e., tuck jump assessment) irrespective of program content; however, the percentage improvement in movement competency was greater when weightlifting progressions were included in the combined resistance training program. Interestingly, the first 14 weeks resulted in significant gains in movement competency and strength, while the second 14 weeks of training resulted in small to moderate improvements in sprint speed and indices of power, which reflect the sequence of strength and movement competency serving as the foundations for gains in power (12, 99).

The benefit of adopting a long-term approach for the athletic development of youth can be seen in a study that showed prepubertal gymnasts were able to make significant improvements in strength, power, speed, and vaulting performance in response to 10 months of neuromuscular training delivered in addition to standard

gymnastics training (160). The study adopted a linear periodized approach to training with a large proportion of time devoted to general preparatory training. The gains in athleticism reported exceeded those of both of a gymnastics-only and a non-gymnastics control group. Interestingly, this study demonstrated a temporal response to training, like the findings of Pichardo and colleagues (187). Significant improvements were observed in different physical attributes over time. After four months, isometric absolute peak force and standing broad jump showed notable increases. By seven months, improvements were seen in countermovement jump height, modified reactive strength index, and drop jump height. Finally, at the 10-month mark, enhancements were noted in peak sprint speed, spring-like behavior, and vaulting vertical takeoff velocity. This study underscores the benefits of long-term engagement in strength and conditioning and that some adaptations take time to materialize, especially in less mature children.

Changes in muscle hypertrophy can significantly contribute to training-induced strength gains in adolescents and adults, yet it is unlikely that muscle hypertrophy is primarily responsible for training-induced strength gains (at least up to 20 weeks) in preadolescents (179, 197). Although some findings do not agree with this suggestion, preadolescents appear to experience more difficulty increasing their muscle mass through a resistance training program due to inadequate levels of circulating hormones (testosterone, growth hormone, insulin-like growth factor). In preadolescent boys and girls, testosterone concentration is between 20 and 60 ng/100 mL; in contrast, during adolescence, testosterone levels in males increase to about 600 ng/100 mL while the levels in females remain unchanged (146).

It appears that preadolescents have more potential for an increase in strength owing to neural factors, such as increases in motor unit activation and synchronization as well as enhanced motor unit recruitment and firing frequency (90, 105, 139, 179, 197). It has also been suggested that intrinsic muscle adaptations,

improvements in motor skill performance, and the coordination of the involved muscle groups could be partly responsible for training-induced strength gains in preadolescents (197). One cannot state without qualification, however, that resistance training does not result in muscle hypertrophy in preadolescents, because it is possible that longer study durations, higher training volumes, and more precise measuring techniques (e.g., computerized imaging) may be needed to uncover potential training-induced muscle hypertrophy in youth who are following a resistance training program. Researchers have reported novel data showing that prepubertal female gymnasts significantly improved muscle thickness (as measured by 2D ultrasound) in addition to measures of isometric peak force and stretch-shortening cycle function following exposure to a 10-month neuromuscular training intervention (161). While muscle thickness is not a direct measure of muscle size or volume, it has been positively correlated with muscle anatomical cross-sectional area (77). The data from Moeskops and colleagues (161) indicate that some change in muscle architecture might be possible even in prepubertal children when they are exposed to long-term neuromuscular interventions.

During and after puberty, training-induced gains in strength are typically associated with gains in muscle hypertrophy and architecture (85, 195) due to hormonal influences and ongoing neural adaptations (86). Although lower levels of testosterone in adolescent females limit the magnitude of training-induced increases in muscle hypertrophy, other hormone and growth factors (e.g., growth hormone and insulin-like growth factor) may be at least partly responsible for their muscle development. [Figure 7.3](#) highlights the factors that contribute to the development of muscular strength—namely, fat-free mass, testosterone concentrations, nervous system development, and the differentiation of fast-twitch and slow-twitch muscle fibers.

Preadolescent boys and girls can significantly improve their strength above and beyond growth and maturation

with resistance training. Neurological factors, as opposed to hypertrophic factors, are primarily responsible for these gains.

In addition to increasing muscular strength, muscular power, and muscular endurance, regular participation in a youth resistance training program has the potential to influence many other health- and fitness-related measures (84, 132, 139). Resistance exercise may favorably alter selected anatomic structures (128, 223), psychosocial parameters (25, 147), and academic outcomes (200) in youth. In addition, ongoing participation in resistance training has been found to reduce injuries in sport and recreational activities (186, 224) and improve motor skills and powerful sporting movements (14, 40, 177).

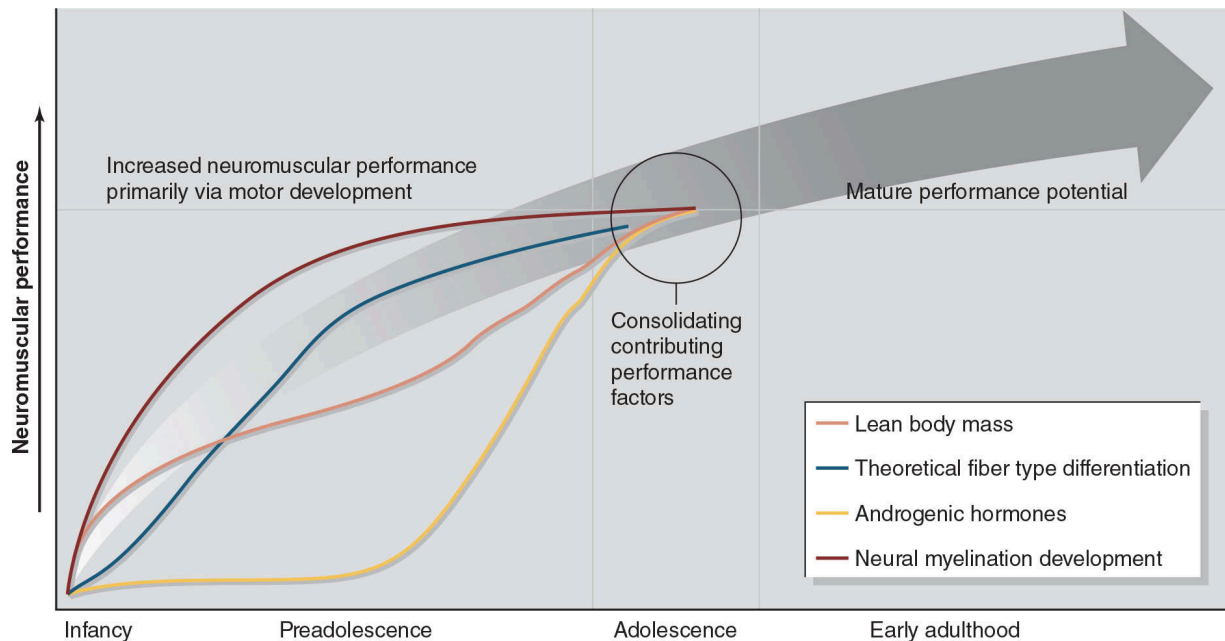


FIGURE 7.3 Theoretical interactive model for the integration of developmental factors related to the potential for muscular strength adaptations and performance.

Reprinted by permission from A.D. Faigenbaum, R.S. Lloyd, and G.D. Myer, GD, "Youth Resistance Training: Past Practices, New Perspectives, and Future Directions," *Pediatric Exercise Science* 25, no. 4 (2013): 591-604.

From a clinical perspective, it has been reported that muscular fitness is associated with lower obesity, insulin resistance, and cardiovascular risk among children and adolescents (43, 84). Of note, meta-analytical data indicate that resistance training can reduce body fat in children and adolescents who are overweight or obese (153). The grades for indicators of youth physical activity around the world are low, which suggests a widespread crisis of physical inactivity (6). Children who are obese or those who lead sedentary lives (e.g., who take a bus to school and watch television or play computer games after school and on weekends) are not ready for 1 to 2 hours of sport training four or five days a week. Current global findings based on a pooled analysis of 3,663 population-representative studies with 222 million children, adolescents, and adults found the burden of obesity has increased and the age of onset of obesity has decreased in most countries (39). Specific to the United States, data from 2013 to 2016 show that 17.8% (95% confidence interval [CI], 16.1%-19.6%) of American youth aged 2 through 19 years were obese and the prevalence of severe obesity was 5.8% (95% CI, 4.8%-6.9%) (174), and the prevalence of many cardiovascular disease risk factors are projected to increase over the next 30 years (118). Although the treatment of childhood obesity is complex, it seems that obese youth enjoy resistance training because it is not aerobically taxing and provides an opportunity for all participants to experience success and feel good about their performance.

In addition to the high prevalence of childhood obesity, other concerning trends in muscular fitness in youth are for those who engage in organized sports or recreational physical activities (66). For example, research indicates that within various cohorts of school-aged children, a range of muscular strength measures (e.g., bent-arm hang and handgrip strength) (51, 206, 209) and motor skill competencies (20, 36, 206) have shown a decrease. Therefore, while all aspiring young athletes will likely benefit from preparatory conditioning inclusive of resistance training, it seems that those who

might benefit the most are the ones who are less fit to begin with. This notion underlines the importance of practitioners understanding and advocating for long-term athletic development as a fundamental concept for all youth (137).

While resistance training does not affect the genotypic maximum, it probably has a favorable influence on growth at any stage of development, as long as appropriate guidelines are followed (70). In fact, regular participation in an exercise program that includes resistance training and weight-bearing physical activities has been shown to enhance bone mineral density in children and adolescents (50, 221). In support of these observations, it has been reported that elite adolescent weightlifters who regularly train with heavy weights while performing multijoint exercises display levels of bone density well above values of age-matched controls (41, 180). These findings may be especially important for girls who are at an increased risk for developing **osteoporosis** later in life, a clinical condition characterized by low bone mass and an increased susceptibility to fractures (231).

Every year, millions of physical activity-related injuries occur during leisure time and sport participation among children and adolescents (23, 196), and it has been estimated that more than 50% of overuse injuries sustained by children could be prevented with simple approaches including coaching education, preparatory conditioning, delayed specialization, adequate time off from organized sport, and proper nutrition and sleep (23, 131, 224). Strength and conditioning professionals can play a pivotal role in preparing young athletes for sport and thereby minimize or offset the incidence and severity of sport-related injuries common to young athletes. In many cases, modern-day youth who enroll in organized sport programs are unfit and ill-prepared to handle the demands of their chosen sport. It has also been suggested that regular participation in a preseason conditioning program that includes resistance training may increase a young athlete's resistance to injury (53, 186, 232). However, given that neuromuscular control in

young soccer players has been shown to decline throughout a competitive season (143) and that injury incidence fluctuates during different phases of the season (e.g., immediately after returning from annual holiday periods) (198, 226), it is prudent for youth to look beyond resistance training as a preseason activity and instead continue to engage in a modified and periodized resistance training program throughout the competitive season and during the postseason and off-season to reduce the risk of subsequent injury.

The goal should be for all children and adolescents to engage in long-term, periodized athletic development programs (inclusive of resistance exercise) rather than short-term programs to enhance physical fitness and reduce the risk of injury.

Early sport specialization is defined as children engaging in intensive, year-round (>8 months) training and/or competition in a single sport, typically beginning during the prepubertal years (131). Traditional early specialization sports included gymnastics, dance, and swimming; however, there is a growing trajectory of youth opting to specialize in other sports at younger ages (21). In many cases, early sport specialization results in young athletes accumulating weekly training volumes that exceed recommended guidelines (21, 202). Even though some coaches argue that early sport specialization is the key to success, this notion has been challenged in the literature (21, 114, 131). Notably, meta-analytical data from 51 studies indicated that while sampling a diverse range of sports led to more measured initial progress, it was associated with more sustainable long-term development toward sporting excellence (96). Research has indicated that youth who participate in a single sport or specialize in a single position at an early age may be at a greater risk of serious injury (114, 150, 238) and overuse injuries (113), developing anterior knee pain (101), possessing less desirable lower limb jump-landing biomechanics (101), and

overtraining and potential burnout (22, 47). Additionally, it seems that late specialization and involvement in a variety of sports and activities during the younger years may be better related to sporting success at a later age (21, 72, 78, 96, 159). Although the total elimination of youth sport injuries is an unrealistic goal, the addition of resistance training as part of a long-term athletic development program may better prepare the young athlete to handle the duration and magnitude of unanticipated forces that develop in practice and game situations (65, 106), whether or not they have specialized. Because of individual differences in growth, maturation, and stress tolerance, resistance training intensity, volume, and rate of progression need to be carefully prescribed since this form of training adds to the chronic repetitive stress placed on the developing musculoskeletal system. Proper progression and program variation will optimize gains, prevent boredom, and reduce the stress that can lead to overtraining. This will be particularly effective when practitioners use reliable data to help inform decision-making around program design, and this evidence-informed approach will also work well with those young athletes who have specialized (163). Additionally, well-planned recovery strategies (e.g., proper cool-down, postexercise meal or snack, and adequate sleep) can help to maximize training adaptations.

Because many sports have a significant strength or power component, it is attractive to assume that resistance training will enhance athletic performance. Though comments from parents, coaches, and children support this contention, scientific reports on this issue are constrained by the ability to comprehensively define and quantify “performance.” However, improvements in selected motor performance skills, such as the long jump, vertical jump, 30-meter (27 yd) sprint, and agility run, have been observed in children who participated in a resistance training program of between 8 and 20 weeks (14, 40, 71, 135, 160, 187). Although direct evaluations of the effects of youth resistance training on sport performance are often based on proxy measures of performance, a progressive

resistance training program will in all likelihood result in some degree of improvement in sport performance in young athletes (57, 89, 102, 133, 134, 210).

Potential Risks and Concerns

Appropriately prescribed youth resistance training programs are relatively safe when compared with other sports and activities in which children and adolescents regularly participate (139, 181). Paradoxically, it seems that the forces placed on the joints of young athletes during sport participation may be far greater, and more difficult to anticipate, than those generated from resistance training programs (65). The belief that resistance training is dangerous for children is not consistent with the needs of children and the documented risks associated with this type of training. Children and adolescents have been injured in the weight room; however, such injuries are more likely to be accidental (98, 169) and typically occur in cases in which the levels of supervision and instruction, technical competency, and training loads are inappropriate (65). This highlights the need for strength and conditioning professionals to heed safety guidelines when working with younger populations. Although epiphyseal plate fractures have been reported in adolescents who were resistance training, these reports were case studies and typically involved the performance of heavy overhead lifts in unsupervised settings (97, 203, 207). An epiphyseal plate fracture has not been reported in any prospective youth resistance training study that adhered to established training guidelines (65). Of note, 1-repetition maximum (1RM) testing (62, 64, 81) and maximal isometric assessments in children and adolescents (48, 162, 167) has proven to be safe, provided that appropriate testing guidelines are followed (i.e., adequate warm-up periods, individual progression of loads, and close supervision of technical competency). If children and adolescents are taught how to resistance train properly and understand resistance training

guidelines and procedures, the risk of an epiphyseal plate fracture is minimal.

Dispelling the Myths of Youth Resistance Training

A wealth of evidence proves children and adolescents can safely and effectively engage in resistance training to promote improvements in physical fitness, reduce the risk of sport-related injury, and enhance measures of health and well-being. However, there still remain certain unfounded myths among some health professionals that may deter youth from engaging in resistance training (67).

- **Resistance training stunts bone growth.**

False! Resistance training stimulates bone remodeling and strengthens bone, muscular tissue, and connective tissue during childhood and adolescence.

- **Resistance training is unsafe for children.**

False! With correct technique, appropriate loading, and high-quality instruction, resistance training is a safe method of training for all youth.

- **Resistance training is only for young athletes.**

False! Resistance training promotes positive physical and psychosocial adaptations in all youth, regardless of whether they participate in competitive sports.

- **Children need to wait until puberty to realize the benefits of resistance training.**

False! Research clearly shows that young children can make meaningful gains in physical fitness following resistance training. Children should be psychologically mature enough to understand and follow instructions.

- **Young girls and boys will become “bulky.”**

False! Studies have shown that resistance training during childhood stimulates neuromuscular adaptations. During adolescence, boys will likely increase muscle mass, but the lower levels of circulating androgens in girls mean they will experience lower levels of absolute muscle growth.

Program Design Considerations for Children

It is important to view resistance training as part of a well-rounded long-term athletic development program for a child that also addresses other fitness goals (e.g., speed, power, balance, coordination, endurance). Although there is no minimal age

requirement for participation in a youth resistance training program, children should have the emotional maturity to receive and follow directions and should want to try this type of activity (139, 215). A pretraining medical examination is not mandatory for apparently healthy children; however, all participants should be screened for any injury or illness that may limit or prevent safe participation in a resistance training program (215). The goals of youth resistance training programs should not be limited to increasing muscular strength but should also include teaching children about their bodies, promoting an interest in physical activity, teaching weight room etiquette, and having fun. It seems likely that children who enjoy participating in physical activities and sport are more likely to be active later in life (220).

Two important areas of concern in the development of youth resistance training programs are the quality of instruction and rate of progression. Strength and conditioning professionals must have a thorough understanding of youth resistance training guidelines, an ability and willingness to demonstrate proper exercise technique, and the requisite pedagogical skills to speak with children at a level they understand (59, 137, 139). Professionals should deemphasize competition between participants and focus on proper technique instead of the amount of weight lifted. Differentiating training sessions can enable practitioners to meet the needs of individual children even when delivering sessions to groups of young athletes. Designing and delivering training programs that pursue a common goal (i.e., movement skill and muscular strength development) but have differentiating levels (e.g., technical difficulty, intensity, range of motion, movement speed) can also be a useful approach to allow for individualization based on the athletes' ability. While some form of variation needs to be applied to achieve ongoing adaptation, this should be applied in a logical and progressive manner, and it is important to keep the program fresh and challenging. Children should be given an opportunity to develop proper form and technique, and there should be a balance between repeated

opportunities to practice and master skills versus introducing new stimuli to keep sessions fun and engaging.

To optimize any training program, practitioners should conduct an initial assessment of the individual's characteristics (i.e., physical, technical, psychosocial) and the demands of the sport or physical activity in which he or she participates, commonly known as a **needs analysis**. With this information collated, practitioners can then adopt an appropriate assessment battery on relevant measures of physical performance and injury risk, in addition to estimations of maturity status. Similarly, consideration should be given to the long-term aspirations of the child prior to finalizing the training program. Of course, it should be recognized that the level of sophistication and rigor of the needs analysis and assessment process is context specific; completing this preliminary work is far easier for a practitioner working one on one with a small number of children in a private facility compared to another teacher or coach working in a high school setting that is responsible for the programming of hundreds of school-aged children. Regardless of the situation, it is important to have some understanding of the current training status of the child to know how to construct the program.

Although only limited data are available regarding the relationship between repetitions and selected percentages of the 1RM in children, it appears that the number of repetitions that can be performed at a given percentage of the 1RM is specific to the given exercise (68). Thus, the minimal strength threshold, when expressed as a percentage of 1RM, may vary between muscle groups, possibly due to the amount of muscle mass involved with each exercise. While practitioners have safely used 1RM (62) and isometric mid-thigh pull testing (160) to determine strength levels in youth within research settings and sport environments, where this is not possible (perhaps owing to time limitations, class size, lack of equipment or coach expertise), alternative means of assessing strength could be used. Strength and conditioning professionals can use predictive equations that estimate 1RM loads from multiple submaximal

repetitions (e.g., 5RM or 10RM) (139). However, strength and conditioning professionals should recognize that such approaches might place the child at heightened risk of injury due to the cumulative fatiguing effects of multiple repetitions on technical form. In order to obtain a surrogate measure of muscular strength without the use of repetition maximum schemes, practitioners could use field-based measures of different jump protocols (e.g., vertical jump or long jump) or handgrip strength, measures that have been significantly correlated to 1RM values in youth (29, 157). Irrespective of the strength testing protocol used, the child or adolescent must be able to demonstrate and maintain correct technical competency throughout the entire test under the watchful eye of a qualified professional.

Advanced multijoint exercises such as the snatch and clean and jerk can be incorporated into a child's program at the appropriate time (i.e., after foundational strength and technique progressions have been completed), but the primary focus, especially early in a resistance training program, should be on developing motor skill competency and muscle strength (168, 194). The **athletic motor skill competencies (AMSC)** have been proposed to represent the foundational movements that underpin all athletic movements ([figure 7.4](#); 194). To develop well-rounded athleticism in youth, practitioners are encouraged to design programs that increase both form (i.e., technical competency) and function (i.e., force-producing capabilities) in each of the AMSC. Practitioners do not necessarily need to expose youth to all of the AMSC in each session, but they should be targeted at various points throughout the training program (194).

Poor technique can put abnormal stress on musculoskeletal tissues and lead to injury. The resistance must be lowered if proper exercise technique cannot be maintained. When learning new exercises, children should start with an unloaded barbell, a long wooden stick, or PVC piping to learn the correct technique. The importance of correct feedback, delivered at the correct time and in

the correct manner, cannot be overestimated during this stage of development to ensure the facilitation of skill development. Youth training should involve ongoing grading and evaluation of the technical performance of different exercises by a trained observer (periodically from formal assessment but consistently during sessions via coach observations), as opposed to simply assessing performance measures (e.g., load lifted or the velocity of movement) (60). Such an approach can be used to continually educate young athletes about proper exercise technique, raise awareness about common technical flaws, and provide coaches and teachers with a tool for assessing learning during practice or physical education. A summary of youth resistance training guidelines is presented in the sidebar.

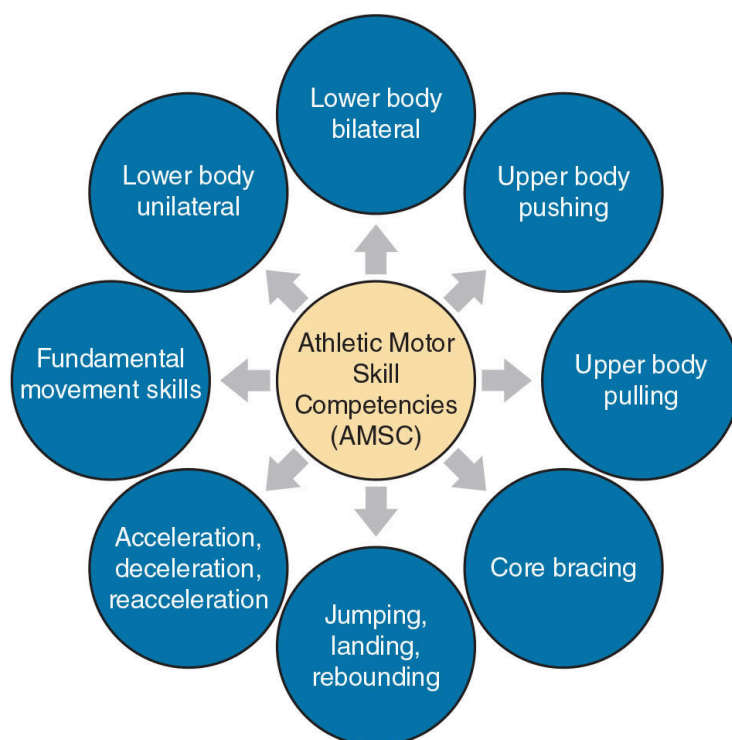


FIGURE 7.4 The spectrum of athletic motor skill competencies (AMSC).

Reprinted by permission from J.M. Radnor, S. Moeskops, S.L. Morris, et al., "Developing Athletic Motor Skill Competencies in Youth," *Strength Conditioning Journal* 42, no. 6 (2020): 54-70.

Youth Resistance Training Guidelines

- Each child should understand the benefits and risks associated with resistance training.
- Competent and caring strength and conditioning professionals should design and supervise training sessions based primarily on the desired training adaptation and technical competency but also with consideration to growth and maturation.
- The exercise environment should be safe and free of hazards, and the equipment should be appropriately sized to fit each child.
- Dynamic warm-up exercises, movement challenges, and opportunities for exploratory play should be performed before resistance training.
- When appropriate, static stretching exercises should be performed after resistance training.
- Carefully monitor each child's tolerance to the exercise stress.
- Begin with light loads to allow appropriate adjustments to be made.
- Increase the resistance gradually (e.g., 5%-10%) as technique and strength improve.
- Depending on individual needs and goals, different set $\times$ repetition combinations (e.g., 3 $\times$ 5) can be performed on a variety of single- and multijoint exercises (and those targeting the strengthening of the core musculature).
- Advanced multijoint exercises, such as the snatch and clean and jerk, may be incorporated into the program, provided that appropriate loads are used and technical proficiency remains a key outcome.
- Two or three nonconsecutive training sessions per week are recommended; however, youth with higher training ages may participate in more frequent resistance training sessions per week.
- When necessary, adult spotters should be nearby to actively assist the child in the event of a failed repetition (e.g., bench press or squat).
- The resistance training program should be systematically periodized throughout the year to ensure that the child or adolescent is exposed to a sequential and varied training stimulus with adequate rest and recovery between training cycles.

Adapted from Faigenbaum et al. (2009) and Lloyd et al. (2014)

Older Adults

The number of men and women over the age of 65 continues to rise, and more opportunities for participation in sports ranging from marathon running to weightlifting are available to older athletes. While the cardiovascular endurance and muscular strength of older competitors or masters athletes are truly exceptional, even the most highly trained athletes experience some decline in performance after age 30 (83, 107). For example, competitive weightlifting ability has been shown to decline with age at a rate of approximately 1% to 1.5% per year until approximately age 70, after which a more dramatic decrease occurs (152). Older individuals who do not engage in physical activity typically see greater decrements in several physical performance measures and increase their risk of debilitating injury (95, 182). Strength and conditioning professionals should understand the physiological changes that occur with aging and the associated trainability of older individuals. In addition, the potential health risks associated with physical activity for older individuals need to be considered.

Age-Related Changes in Musculoskeletal Health

Significant changes in body composition with advancing age can lead to the development of physical functional impairments and injury. The well-documented loss of bone and muscle with age not only makes activities of daily life such as getting out of a chair and opening a window more difficult but also increases the risk for falls, fractures, and long-term disability (37, 95). Bones become fragile with age because of a decrease in bone mineral content that causes an increase in bone porosity. The bone mineral content and microarchitecture of bone can deteriorate to such an extent that the risk of bone fracture, particularly of the hip, spine, or wrist during a fall, is increased (2, 76). **Osteopenia** is defined by a bone mineral

density between -1 and -2.5 standard deviations (SD) of the young adult mean, whereas osteoporosis is defined by a bone mineral density below -2.5 SD of the young adult mean (119). These conditions, which result in bones with less density and strength, are serious concerns for older people (particularly women) because they heighten the risk of skeletal fracture and poor bone health (2). The slow but progressive loss of bone with age has been linked to physical inactivity and to hormonal, nutritional, mechanical, and genetic factors (7, 11). [Figure 7.5](#) shows the structural difference between normal, healthy bone and bone at the stage of osteoporosis.

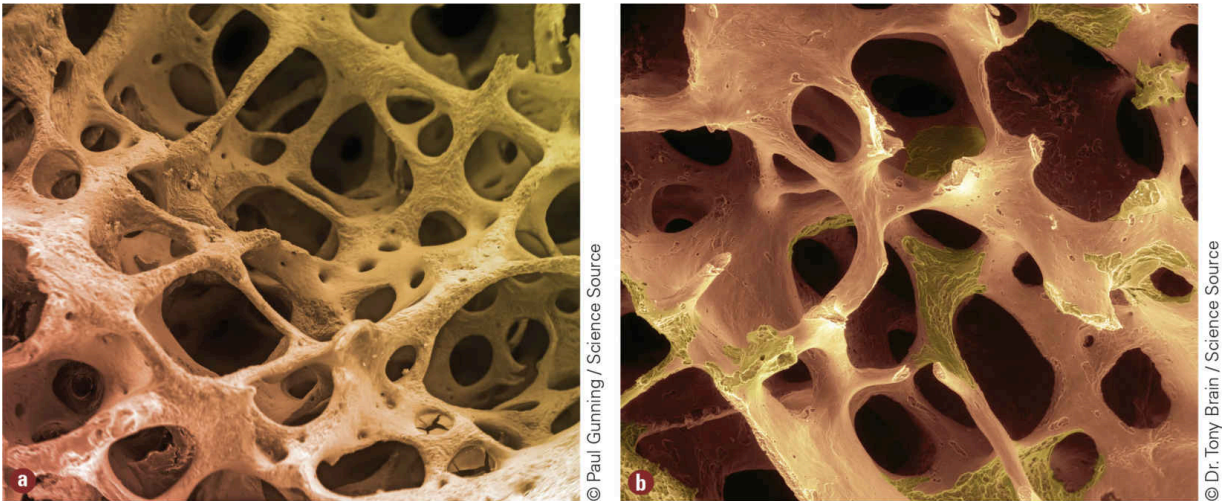


FIGURE 7.5 Structural differences between (a) normal, healthy bone, and bone at the stage of (b) osteoporosis.

© Paul Gunning / Science Source

Advancing age is also associated with a loss of muscle mass, strength, and function (19, 38). The term **sarcopenia** refers to a progressive and generalized skeletal muscle disorder that is associated with an increased likelihood of adverse outcomes including falls, fractures, physical disability, and mortality (42). Computed tomography has revealed that after age 30 there is a decrease in the cross-sectional areas of individual muscles, along with a decrease in muscle density, reductions in tendon compliance,

and an increase in intramuscular fat (2, 110). These changes seem to be a predictable consequence of advancing age and seem to be most pronounced in women (110). The observed muscle atrophy with aging appears to result from physical inactivity and a gradual and selective denervation of type II muscle fibers (136, 192, 237). Decreased muscle mass results in a loss of muscle strength. In one report, 40% of women aged 55 to 64 years, 45% of women aged 65 to 74 years, and 65% of women aged 75 to 84 years were unable to lift about 10 pounds (4.5 kg) (115). The reduction in the size and number and the gradual denervation of muscle fibers (100, 192) and the age-related changes in motor unit morphology and properties (108, 204) also lead to a decrease in the ability of a muscle to generate power (i.e., exert force rapidly); in fact, power recedes at a faster rate than muscle strength with aging (79). Since everyday activities require a certain degree of power production, a decrease in the ability of muscles to produce force rapidly may adversely affect the ability of older adults to safely perform activities such as stair climbing and walking. Factors that may contribute to the age-related decline in muscle strength and power include reductions in muscle mass, nervous system changes, hormonal changes, poor nutrition, and physical inactivity (16, 112). The functional consequences of these age-related changes are significant because the magnitude and rate of change influence the age at which a person may become functionally dependent (e.g., unable to perform household tasks or rise from a chair) or reach a threshold of disability. A summary of adaptations to aging and resistance training is presented in [table 7.1](#).

TABLE 7.1 Summary of the Effects of Aging and Resistance Training

Physical or physiological variable	Effect of aging	Effect of resistance training
Muscular strength	Decreases	Increases
Muscular power	Decreases	Increases

Physical or physiological variable	Effect of aging	Effect of resistance training
Muscular endurance	Decreases	Increases
Muscle mass	Decreases	Increases
Muscle fiber size	Decreases	Increases
Muscular metabolic capacity	Decreases	Increases
Resting metabolic rate	Decreases	Increases
Body fat	Increases	Decreases
Bone mineral density	Decreases	Increases
Physical function	Decreases	Increases

Advancing age is associated with a loss of muscle power, muscle strength, and muscle mass, which is largely attributable to physical inactivity. A direct result of these losses is a decrease in functional independence and an increased risk of falls and fractures.

Age-Related Changes in Neuromuscular Function

Seniors are at an increased risk of falling, which can lead to serious health, psychosocial, and economic consequences that negatively affect overall quality of life. Specifically, falls can result in pain syndromes, joint dislocations, skeletal fractures, limitations on daily functional activities, and a reduction in self-confidence (95, 211). Falls can also lead to permanent disability, institutionalization, and fatalities (46). Intrinsic factors that lead to increased risk of falls in seniors include decrements in muscle strength and power (190), reaction time (2), and impaired balance and postural stability (205). Muscle activity before (**preactivation**) and immediately following (**cocontraction**) contact with the ground is an important mediator of braking and dynamic stabilization in both young and old individuals. Increased preactivation helps increase stiffness of the

limb using fast stretch reflexes to better prepare the limb for ground contact. Cocontraction is a motor control strategy that dynamically stabilizes the joint; however, due to the simultaneous activation of both the agonist and antagonist muscle groups crossing the same joint, net joint moments and agonistic force outputs are reduced (236). Research shows that seniors rely on increased levels of muscle cocontraction as a compensatory mechanism to offset their increased balance difficulties and to minimize postural sway (171). Research also highlights that reduced mechanical power output when walking (especially during the push-off phase) of is one of the biomechanical aging features of gait (45). Intuitively, this literature would suggest that older adults should use a variety of training modes specifically designed to offset these natural reductions in neuromuscular function. Such methods include low-intensity plyometrics, strength training, balance and dynamic stabilization exercises, and proprioception training to develop the ability to react more efficiently with the ground (92, 95, 189).

Physical activity interventions can be effective in improving neuromuscular function and preventing falls in seniors (95, 112, 222). However, simply increasing physical activity will not prevent falls on its own. Rather, seniors must engage with, and adhere to, multidimensional, periodized programs that incorporate elements of both resistance and balance training (76, 112, 222). Furthermore, current guidelines suggest that as with any population, training programs for seniors should overload progressively to promote a challenging training environment, and that training should be completed on a frequent basis to provide individuals with a sufficient training dosage (76). Of note, resistance training as a stand-alone training method does not seem to prevent the risk of falls (211), and balance and flexibility training must accompany resistance training in order to provide the requisite training stimulus to reduce fall risk.

Aerobic exercise should also be featured in a well-balanced exercise program for older adults, because data indicate that concurrent strength and aerobic endurance training can be delivered

together to improve multiple measures of physical fitness (8). Notably, maximal aerobic power decreases with age in adults as a consequence of various physiological changes that accompany aging; for example, muscle mass and strength decreased (130) and fat mass increases (129), with men showing a more rapid decline than women (233). On average, when women and men are matched by age, the aerobic power values of women range from 73% to 85% of the values of men (229); however, the general physiological response to training in later life is equivocal between men and women (4, 183). The differences in aerobic power may be caused by several factors, including women's higher percentage of body fat and lower blood hemoglobin values and men's larger heart size and blood volume (24, 34). Notwithstanding the importance of a varied training program, the relative importance and potency of resistance training in increasing muscular strength, muscular power, and bone mineral density, while counteracting the age-related declines in contractile function and muscle atrophy, in addition to its many other health benefits for seniors, cannot be overlooked (76).

Resistance Training for Older Adults

Aging does not appear to enhance or reduce the ability of the musculoskeletal system to adapt to resistance exercise, and as a training mode, resistance training can be an efficacious tool against the declines associated with sarcopenia (28). Significant improvements in muscular strength, muscular power, muscle mass, bone mineral density, and functional capabilities (e.g., gait speed) have been observed in older people who participated in progressive resistance training programs (76, 112, 145). For older adults, such improvements enhance exercise performance, decrease the risk for injury, promote independent living, and improve quality of life. Because of the age-related changes in musculoskeletal health, resistance exercise is a beneficial mode of training for older populations who need to enhance musculoskeletal strength and power and resist decrements in muscle mass, bone mineral density,

and functional abilities. Resistance training can also improve mobility and enhance physical functioning that enable seniors to perform **activities of daily living (ADL)** and preserve independence (76). Furthermore, data also show that muscular strength is an important factor for reducing the risk of cardiovascular disease, type 2 diabetes, cancer mortality, and all-cause mortality (87, 164).

Resistance training provides a highly effective means of attenuating physiological changes associated with aging, improving physical function, and decreasing the risk of chronic health conditions.

Responsiveness to Resistance Training in Older Adults

A great deal of attention has focused on strategies to improve the musculoskeletal health of older men and women. Because of the deconditioned state of many seniors, desirable changes in muscle strength and function can result from a variety of resistance training protocols, particularly during the first few weeks of training (82). Previously sedentary older men have more than doubled knee extensor strength and tripled knee flexor strength following a 12-week resistance training program (80), and similar observations have been made in older women after 12 weeks of resistance training (33). In a seminal study, the ability of very old men and women (87-96 years old) to improve muscular strength was demonstrated following only eight weeks of resistance training (73). Meta-analytical data has indicated that in a sample of more than 800 seniors aged 75 to 97 years of age, resistance training significantly increased measures of muscle strength and increases in whole-muscle hypertrophy (94). Meta-analytical data also show that while sex differences may exist in resistance training adaptations in seniors when data are considered on an absolute basis with males making larger gains than females, these sex differences are removed, or in some measures reversed, when data are interpreted on a relative

scale (117). Collectively, resistance training appears to be an effective means to mitigate age-related **dynapenia** (76).

Improvements in gait speed, stair-climbing ability, balance, and overall spontaneous activity have also been associated with training-induced strength gains in older populations (145). Declines in muscle power are associated with disability and the ability to perform ADL, particularly in females (112), and the need to minimize age-related deficits in muscle power is an important part of exercise prescription for seniors (76, 112). Evidence indicates that resistance training specific to power development may help optimize functional abilities in older adults (2, 92, 178) and that power training may be as effective as traditional resistance training in developing muscle architecture and neuromuscular activation properties of the lower limb (230). In some instances, it has been suggested that high-velocity power training has a greater influence on the ability to produce explosive force than traditional progressive resistance training (92, 191). For example, Fielding and colleagues (74) showed that older adults who completed high-velocity resistance training made greater gains in peak power and similar gains in maximum strength compared to seniors who trained at slower velocities over a 16-week training period. Similar results were reported by Reid and colleagues (199); in that study, older adults who completed a high-velocity, high-power resistance training program made significantly greater gains in leg press peak power than individuals who followed a slow-velocity, progressive resistance training program. While the optimal training protocol for improving muscular strength and power in seniors is not known, it appears a dose–response relationship exists between training intensity and improvements in muscular strength and power (44), with higher-intensity resistance training being more effective in developing maximal muscle strength than moderate- or low-intensity training (9, 214).

Regular participation in a resistance training program also seems to have profound anabolic effects in older populations (33, 73, 80). Computed tomography and muscle biopsy analysis showed evidence

of muscle hypertrophy in older men who participated in a high-intensity resistance training program (80), and other investigations involving older adults have shown that resistance training can improve nitrogen retention, which can have a positive effect on muscle protein metabolism (27, 239). Meta-analytical data have indicated that resistance training is also an effective treatment to improve body fat mass and for muscle strength and performance in seniors diagnosed with sarcopenia (35). Resistance training has also been shown to have an important effect on energy balance in older adults, as evidenced by an increase in the resting metabolic rate of men and women who resistance train (26). It is noteworthy that dietary modifications (a change in total food intake or selected nutrients) in addition to resistance training promotes a greater hypertrophic response than resistance training alone in older men (154).

Although the response of bone to resistance exercise is influenced by a complex interaction of many variables (e.g., hormonal status, activity history, and nutrition), it has been reported that resistance exercise has a positive effect on bone health in older men and women (103, 173). Regularly performed resistance training can offset the age-related declines in bone health by maintaining or increasing **bone mineral density (BMD)**. Notably, meta-analytical data on 1,130 participants (mean age ~70 yrs) indicate that progressive resistance training programs stimulated significant increases in muscle strength and BMD of the femur/hip, with adaptations in both strength and BMD favoring lower volumes and higher external loads (173). Furthermore, a systematic review of 25 studies indicated that while ≥ 2 sessions per week of high-velocity resistance training can significantly increase BMD, skeletal adaptations can be lost when resistance training is stopped for >6 months (103). Researchers have also documented that some strength, power, and fiber type adaptations in older men are only partially preserved after just 12 weeks of detraining; however, strength gains can be recovered following a period of retraining (18).

Collectively, these studies show the transient nature of neuromuscular adaptations and underline the importance of viewing resistance training as a lifelong commitment. Resistance training may also reduce the risk of osteoporotic fractures by improving dynamic balance, muscle mass, and overall level of physical activity (2). While resistance exercise undoubtedly can improve bone health in older adults, the interaction of exercise with hormonal and nutritional factors influences the degree of benefit of the exercise program. Furthermore, bones retain the beneficial effects of exercise only as long as training is continued. During periods of inactivity, bone density tends to revert to preexercise levels (111).

Older men and women can make significant improvements in strength, power, mobility, and functional ability in response to periodized and progressive resistance training. Aerobic, resistance, and balance exercise are all beneficial for older adults, but only resistance training can increase muscular strength, muscular power, and muscle mass.

Program Design Considerations for Older Adults

Resistance training is currently recognized as an important component of a well-rounded fitness program for older adults (2, 76, 112). Because age-related losses of musculoskeletal strength, power, and mass may be almost universal, programs designed to maintain or improve musculoskeletal health in older adults should be implemented. Not only can regular participation in a resistance training program offset some of these age-related losses, but it can help older people maintain a more active, higher-quality lifestyle and promote psychological and cognitive well-being (76, 112).

The fundamental principles of designing a resistance training program for an older person and a younger person are basically the same, but strength and conditioning professionals need to consider several concerns when working with seniors. Attention should be

given to preexisting medical ailments, prior training history, and nutritional status before the beginning of a resistance training program that may predispose seniors to increased risk of exercise-induced injury or illness. Even though older populations retain the capacity to adapt to increased levels of physical activity, safe and effective exercise guidelines must be followed.

Before participation in an exercise program, seniors should complete a medical history and risk factor questionnaire (76). Potential limitations and possible restrictions for physical activity can be ascertained from this information. In some cases, physician clearance is required before the initiation of a moderate or vigorous exercise program (76)—for example, in cardiac rehabilitation patients or cancer survivors. Any questions regarding a participant's medical status (e.g., heart disease, hypertension, arthritis, osteoporosis, or diabetes mellitus) should be answered by a health professional. After this information has been obtained, a preprogram evaluation to document baseline measurements and assess responses to specific exercise modalities should be performed. Although a treadmill exercise test is often used to evaluate cardiovascular responses to aerobic exercise, a strength test (preferably on the equipment used in training) should be performed to assess responses to resistance exercise and aid in the exercise prescription. Of note, while resistance machines may be used in the early stages of a training program with older adults due to balance and flexibility limitations, where appropriate, older adults should sensibly incorporate free weight, multijoint resistance training exercises into the program to provide a greater overall training stimulus and place a greater demand on postural stability. Various methods of assessing muscular strength, including repetition maximum testing, can be reliably used in senior populations, provided that appropriate testing guidelines are followed (76). Strength and conditioning professionals should be aware of the potential risks of breath holding (Valsalva maneuver) for older adults (76). While the Valsalva maneuver assists in stabilizing the trunk and

vertebral column during performance of various resistance training exercises, owing to the sudden rise in systolic and diastolic blood pressure that the technique creates, this technique is generally discouraged in seniors (1, 149), especially in those individuals with absolute and relative contraindications (e.g., those with coronary heart disease, uncontrolled hypertension, musculoskeletal limitations, and pacemakers or defibrillators) (76).

Evidence suggests that resistance training can be safe for seniors when individuals adhere to appropriate training guidelines (2, 76, 112). On the other hand, poorly designed programs can be potentially hazardous. For example, failing to provide sufficient rest periods between sets and between different exercises, programming exercises that are technically too challenging, or increasing the intensity of exercise (typically the external load against which the individual is working) all increase the likelihood of resistance training-related injury. Like the situation with youth populations, the resistance training stimulus should not be increased at the expense of technical competency for older adults.

Untrained seniors who begin resistance training should start at a relatively low exercise intensity and volume, and the exercise prescription should be individualized (76). Notably, meta-analytical data have indicated that while single sets provided adequate stimulus for adaptations in upper body strength, multiple sets produced greater improvements in lower-limb muscle strength and muscle quality in older adults (148). Although higher intensities and volumes can be tolerated by some older men and women who have resistance training experience, the early phase of the training program should be directed toward emphasizing the benefits of resistance training, addressing individual needs, building confidence, and learning proper exercise technique while minimizing the potential for muscle soreness and injury. Less intense training during the first few weeks of an exercise program may also be beneficial for seniors who are apprehensive about participating in a resistance training program. Following the initial adaptation period, the training

program can gradually progress if it continues to meet the needs and medical concerns of each person. In training older men and women, it is particularly important to focus on the interaction of major muscle groups used in everyday activities, such as load carrying and climbing stairs.

During the early stages of a resistance training program for older adults, investing in education, reducing anxieties, and building confidence are key outcomes to promote long-term engagement.

Once participants master the performance of basic resistance exercises, more advanced and demanding exercises such as standing postures with free weights (barbells and dumbbells), multidirectional medicine ball exercises, and advanced balance training (e.g., single-leg stands and circle turns) can be incorporated into the program. Seniors should gradually progress from 1 set of 8 to 12 repetitions at a relatively low intensity (e.g., 40%-50% 1RM) to higher training volumes and intensities (e.g., 3 sets per exercise with 60%-80% of 1RM), depending on individual needs, goals, and abilities (76, 82). The current consensus indicates that for each of the primary muscle groups, older adults should progress toward 2 or 3 sets of 6 to 12 repetitions of one or two multijoint exercises, working at 70% to 85% of the 1RM, two or three times per week (76). In addition, high-velocity power exercises can be gradually incorporated into the overall training regimen if seniors have successfully completed a general resistance training program. Current recommendations for increasing power in healthy older adults include the performance of 1 to 3 sets per exercise with a light to moderate load (40%-60% 1RM) in concentric movements for 6 to 10 repetitions with high repetition velocity (76). However, plyometrics can also serve as an effective means to develop high rate-of-force development in various expressions of the stretch-shortening cycle in older adults when properly programmed in supervised settings (227).

Dispelling the Fears of Resistance Training for Older Adults

Despite the publicized benefits of resistance exercise for older adults, misperceptions remain about the notion of lifting weights. Literature has noted the phenomenon of a “fear of resistance exercise,” which can lead to older adults demonstrating avoidance behavior to resistance exercise due to apprehension, anxiety, or a lack of confidence (4). Similarly, a lack of awareness or understanding can be a barrier to older adults participating in resistance exercise, which underlines the importance of correct education and the promotion of individualized approaches (31, 109). To dispel the fears of resistance training, certain key messages should be promoted to older populations:

- **Resistance training can cause injury to the musculoskeletal system in older adults.**

False! When programs are technique focused, individualized, and designed and delivered by appropriately qualified professionals, resistance training is a safe and effective mode of training for older adults and can promote positive adaptations in both muscle and bone (76).

- **Resistance training is not safe for older adults with cardiovascular or musculoskeletal contraindications.**

False! Resistance training can easily be modified to accommodate for such contraindications. Initially training at lower volumes and intensities, avoiding the Valsalva maneuver, reducing ranges of motion, and simplifying exercise variants are some of the ways in which prescription can account for contraindications and help develop confidence in older adults (76).

- **Resistance training takes too much time.**

False! Improvement in muscle strength and function can be achieved from exposures to a minimal dose program (e.g., ≤ 60 mins/session, 2 days/week) (75). Exercise “snacking,” defined as multiple short bouts (e.g., < 10 mins) of muscle strengthening exercise, has also been shown to promote adaptations in basic function in older adults (185, 234).

- **Resistance training requires access to a fitness center.**

False! Improvements in both health- and skill-related components of fitness have been realized by older adults following exposure to home-based resistance exercise. Notably, supervised training is preferable; however, home-based muscle strengthening activities can provide some form of meaningful adaptation (32).

- **Resistance training will make older adults too muscular.**

False! Appropriately prescribed resistance training can stimulate gains in muscle strength without significantly adding muscle bulk. Maintaining or increasing muscle

mass, strength, and power are important for performing ADLs and physical independence.

A resistance training program for older men and women should vary in volume and intensity throughout the year to lessen the likelihood of overtraining and ensure that progress is made throughout the training period. Because recovery from a training session may take longer in older populations, a training frequency of two or three sessions per week is recommended, at least during the initial adaptation period (76, 112). Strength and conditioning professionals should be sensitive to the concerns of individuals and be able to modify a training program based on the person's health history and individual goals. With competent instruction and support from friends, older men and women can gain confidence in their ability to resistance train, which may be enough to ensure good adherence to the program. However, since a majority of older adults do not currently engage in resistance training activities (54, 122, 213), professionals may initially need to increase awareness of the fitness benefits associated with resistance exercise, address the concerns that seniors may have about participating in a resistance training program, and adapt the exercises and intensity to the needs of the participants.

Safety Recommendations for Resistance Training for Older Adults

- Older adult participants should be prescreened, since many older people suffer from a variety of age-related medical conditions. If necessary, medical advice should be sought concerning the most appropriate type of activity.
- Participants should warm up for 5 to 10 minutes before each exercise session. An acceptable warm-up includes low- to moderate-intensity aerobic activity and calisthenics.
- Older adults should perform static stretching exercises either before or after—or both before and after—each resistance training session.

- Older adults should use a resistance intensity that does not excessively overload the musculoskeletal system.
- Participants should avoid performing the Valsalva maneuver during resistance training to avoid an abnormal increase in blood pressure.
- Older adults should be allowed 48 to 72 hours of recovery between exercise sessions.
- They should perform all exercises within a range of motion that is pain free.
- As with any individual engaging with resistance training, older adults should receive individualized exercise instruction from qualified instructors.

Researchers have reported that social support, access to qualified fitness professionals, improving the ability to complete daily activities, and decreasing the fear and likelihood of falls are some of the motivators for older adults consistently engaging in resistance and balance exercise.

An additional consideration related to resistance training in older men and women is appropriate nutrition. The quality and quantity of a person's food intake (or perhaps selected nutrients) may mean the difference between losing and gaining muscle mass. It seems that adequate amounts of protein are essential for muscle hypertrophy in older people (126, 151). Furthermore, inadequate intakes of macronutrients (fat, protein, and carbohydrate) and micronutrients (vitamins and minerals) are associated with potential negative health consequences, including fatigue, compromised immune function, and delayed recovery from injury. Improving an older person's food intake not only improves health but may also optimize adaptations to resistance training.

Conclusion

Based upon the available research, resistance training can be a safe and effective method of conditioning for males and females of all ages and abilities. The potential benefits are multifactorial, including positive effects on a variety of physical performance variables (e.g., strength and power), health markers (e.g., body composition and cardiac function), and psychosocial development (e.g., self-image, confidence, well-being). Moreover, regular participation in a resistance training program can reduce the risk of sport- and physical activity-related injuries in athletes and nonathletes of all ages and abilities and promote independent living in seniors. Though the fundamental principles of resistance training are similar for people of all ages, there are unique concerns specific to youth and senior populations. Knowledge of age- and maturity-related differences is essential to the development and evaluation of safe and effective resistance training programs. Strength and conditioning professionals should be aware that individual responses to resistance exercise may vary greatly and should be sensitive to the individual needs of all participants.

Over the past few decades, coaches, clinicians, and exercise scientists have added to our understanding of age- and maturity-related differences and their implications for resistance exercise. Collectively, this work has shown the beneficial effects of resistance training on individuals of all ages and has provided the foundation for recommendations about the design of strength and conditioning programs. The information in this chapter and other chapters should help strength and conditioning professionals understand and appreciate age- and maturity-related differences and enhance their ability to develop safe and effective resistance training programs for children, adolescents, and older adults alike.

KEY TERMS

activities of daily living (ADL)
adolescence
adulthood
apophyseal
athleticism
athletic motor skill competencies (AMSC)
biological maturation
bone mineral density
childhood
chronological age
cocontraction
development
diaphysis
dynapenia
epiphyseal plate
growth
growth cartilage
long-term athletic development
maturation
needs analysis
older
osteopenia
osteoporosis
peak height velocity (PHV)
preactivation
puberty
resistance exercise
sarcopenia
senior
technical competency
training age
young athlete
youth

STUDY QUESTIONS

1. In reference to biological maturation, the acronym PHV stands for which of the following?
 - a. pediatric height value
 - b. peak height velocity
 - c. partial height value
 - d. peak height value
2. During the developmental years, the primary ossification center where most bone formation occurs is referred to as which of the following?
 - a. diaphysis
 - b. growth plate
 - c. head of the bone
 - d. uniphysis
3. Improvements in muscle strength and power during childhood are predominantly due to adaptations in which system of the body?
 - a. skeletal system
 - b. circulatory system
 - c. neuromuscular system
 - d. immune system
4. Which of the following is the progressive and generalized skeletal muscle disorder that is associated with an increased likelihood of adverse outcomes including falls, fractures, physical disability, and mortality in older adults?
 - a. hyperplasia
 - b. hypertrophy
 - c. neuropathy
 - d. sarcopenia
5. What condition is deemed to be present when an individual's bone mineral density is 2.5 standard deviations (SD) or more below the average of the young adult mean?

- a. osteoclast
- b. osteopenia
- c. osteoporosis
- d. osteoblast